

**UNIVERSITY OF MYSORE
MYSORE-570005**



A
Project Report On

**REAL-TIME VEHICLE DETECTION IN INTELLIGENT
TRANSPORT SYSYTEM**

Submitted in partial fulfillment of the requirements for the award of the
degree of
BACHELOR OF COMPUTER APPLICATION

Submitted by

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CERTIFICATE

Certified that the project work “**REAL-TIME VEHICLE DETECTION IN INTELLIGENT TRANSPORT SYSYTEM**” is a Benefited work carried out by a team of **YASHAWANTH K S, MANOJ M S, ADARSH S N, SUKITH GOWDA V** , REG NO: U01CT22S0018, 0077, 0162, 0168 in fulfilment of requirements for the award of Degree in **BACHELOR OF COMPUTER APPLICATION** by University of Mysore, Mysore During the academic year 2024-2025. It is certified that all corrections and suggestions Indicated for the Internal Assessment have been incorporated in the Report deposited in the Department library. The project report has been approved as it satisfies the academic Requirements in respect of project work prescribed for the said degree.

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*We are the student of Final Year BCA of Seshadripuram Degree College, Mysuru, hereby declare that the dissertation entitled **“REAL-TIME VEHICLE DETECTION IN INTELLIGENT TRANSPORT SYSYTEM”** has been carried out by us and submitted in partial fulfilment of the requirement for the award of **Bachelor of Computer Application** affiliated to the University of Mysore, Mysuru during the academic year 2024-2025. Further the matter embodied in the report is an Original and Bonafide work done by us.*

To my knowledge this dissertation as not been submitted to any other college or Universities at any time till this.

Place: Mysuru

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ACKNOWLEDGEMENT

*We take immense pleasure in thanking **Dr. Sowmya Erappa K**, Principal of Seshadripuram Degree College and our internal guide **Mr. Tejaswi S** Assistant Professor in Department of Computer Science, Seshadripuram Degree College, for his guidance and excellent suggestions, which helped us in completing the project work entitled **“REAL-TIME VEHICLE DETECTION IN INTELLIGENT TRANSPORT SYSTEM”**.*

ABSTRACT

This survey examines the evolution and current state of vehicle detection algorithms with a particular focus on deep learning techniques, especially the YOLO (You Only Look Once) family of models. We analyse the transition from traditional methods based on manual feature extraction to modern deep learning approaches that offer superior performance in real-world environments. The survey covers key challenges in vehicle detection including environmental variations, occlusions, and computational constraints, while highlighting recent advancements that address these limitations. Special attention is given to variants of YOLO models and their applications in intelligent transportation systems (ITS), autonomous driving, and traffic monitoring. We identify research gaps, particularly regarding model performance under diverse real-world conditions, and suggest directions for future research to enhance the robustness and efficiency of vehicle detection systems.

Index Terms-You Only Look Once (YOLO), Intelligent Transportation System (ITS), Vehicle Detection, Deep Learning.

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CHAPTER 1

INTRODUCTION

1.1 PROJECT DESCRIPTION

The rapid urbanization and increasing vehicle density in modern cities have created significant challenges for traffic management and urban mobility. As cities strive to become "smarter," there is a growing need for efficient and accurate vehicle detection systems that can provide real-time information for traffic monitoring, congestion management, and intelligent transportation systems (ITS). Vehicle detection serves as the fundamental building block for numerous applications including traffic flow analysis, parking management, autonomous driving, and license plate recognition. Early approaches to vehicle detection relied on manual feature extraction techniques such as background subtraction, frame differencing, and optical flow analysis. While these methods provided initial capabilities, they faced significant limitations in real-world scenarios due to their sensitivity to environmental factors like lighting conditions, weather variations, and object deformations. Additionally, these traditional methods often struggled with computational efficiency, making real-time detection challenging.

The advent of deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the field of vehicle detection. The transition from hand-crafted features to data-driven feature engineering has significantly improved detection accuracy and robustness in complex traffic environments. Among various deep learning architectures, the YOLO (You Only Look Once) family of models has gained considerable attention due to its balance between detection speed and accuracy, making it particularly suitable for real-time applications.

YOLO approaches vehicle detection as a regression problem, processing the entire image in a single pass to simultaneously predict bounding boxes and class probabilities. This one-stage detection methodology contrasts with two-stage detectors like Faster R-CNN, offering computational efficiency crucial for embedded systems and real-time applications. The evolution of YOLO through various versions has progressively addressed limitations such as small object detection, localization errors, and environmental robustness.

CHAPTER 2

LITERATURE SURVEY

Early vehicle detection algorithms relied on manual feature extraction, which involved methods like background update, frame difference, and optical flow. These methods often struggled with real-time detection due to high computational demands and sensitivity to environmental factors such as lighting and object deformation. YOLO (You Only Look Once) and SSD (Single-Shot MultiBox Detector) represent one-stage detection methods that predict bounding boxes and classifications directly from the entire image. YOLO, in particular, is noted for its speed, achieving up to 155 fps, but faced challenges with small object detection. Recent studies emphasize the importance of high-dimensional datasets for training deep learning models. There is a noted need for large-scale benchmark datasets to enhance the performance and reliability of these models. The literature review indicates that most existing solutions use simpler classification approaches rather than advanced object detection techniques.[1]

The paper argues that vehicle detection can provide more comprehensive data, such as road congestion and vehicle interactions, which are not captured by mere classification methods [4]. The literature review also discusses the application of artificial intelligence algorithms, ranging from simple multilayer perceptrons to more complex networks like Faster R-CNN and YOLO. The paper specifically mentions the use of YOLO v3 for vehicle detection in their own study, indicating a shift towards more advanced AI techniques in speed estimation [5]. The paper highlights the integration of remote sensing, photogrammetry, and deep learning algorithms, which has led to significant advancements in real-time and accurate vehicle detection. This integration is crucial for applications in urban management, especially in traffic and parking lot management [6]. The proposed YOLOv5n-L algorithm aims to address these challenges by integrating depth wise separable convolutions and a C3Ghost module to reduce model parameters and enhance detection speed. Additionally, it incorporates a Squeeze-and-Excitation attention mechanism to improve accuracy and mitigate environmental interference [3]. Early research relied on traditional machine learning approaches, which used hand-crafted features—such as edges, color, texture, and shape—and classifiers like SVM and Random Forest to detect vehicles. However, these methods struggled with unseen images and the inherent variability of aerial views. The advent of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized the field with two-stage detectors like Faster R-CNN offering high accuracy, and one-stage detectors like YOLO and SSD providing the speed required for real-time applications [8].

Traditional methods relied on handcrafted features and classical machine learning algorithms, such as SVM and Random Forest, which had limitations in complex traffic environments. The introduction of deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized the field by improving accuracy and robustness in real-world scenarios. Recent developments focus on real-time detection using architectures like YOLO (You Only Look Once), Faster R-CNN, and SSD. The survey highlights the challenges of detecting vehicles in varying conditions, including occlusions, different weather conditions, and dense traffic. Additionally, improvements such as transfer learning and feature fusion techniques have been

incorporated to enhance detection accuracy. The study emphasizes the importance of optimizing models for real-time applications in intelligent transportation systems (ITS), where speed and accuracy are crucial. Overall, the literature survey provides a comprehensive overview of existing techniques and sets the foundation for the proposed YOLO-based approach to vehicle detection [9].

The paper introduces the You Only Look Once (YOLO) algorithm, which is built on the AlexNet architecture. The evolution of YOLO through its versions (YOLOv3, YOLOv4, and YOLOv5) is also discussed, emphasizing its effectiveness in object detection tasks [10]. The architecture of YOLO v4 incorporates various improvements over its predecessors, which enhances its ability to learn features from images. [11]. Real-time object detection is crucial for Advanced Driver Assistance Systems (ADAS) and decision-making in autonomous vehicles (AVs). It provides essential information that enhances the safety and efficiency of AV systems [12]. Vehicle detection is crucial for various applications, including parking lot management, vehicle tracking, and license plate recognition. It serves as the foundation for these technologies, emphasizing its significance in the field of computer vision and autonomous systems [14]. The paper "M-YOLO: A Nighttime Vehicle Detection Method Combining Mobilenet v2 and YOLO v3" presents a comprehensive overview of existing vehicle detection methods, particularly focusing on nighttime scenarios. Here are the key points from the literature survey.[15]. The paper discusses the development of various CNN models for object detection, noting that many of these models struggle with real-time analysis due to slow recognition speeds. YOLO is highlighted as a leading model that balances detection precision and speed, although it still has limitations, such as processing each image separately, which can lead to missed identifications in continuous video streams [17]. The paper emphasizes the significance of ITS in urban environments, highlighting its role in reducing fuel consumption, monitoring emissions, and optimizing traffic flow. These systems are crucial for addressing environmental and economic challenges associated with traffic congestion [18]. The survey specifically mentions the YOLO (You Only Look Once) family of models, which are known for their speed and efficiency in real-time object detection. The paper indicates that while YOLO models have shown promise, they still face challenges in scenarios involving occluded and densely packed vehicles, particularly under varying weather conditions [19].

CHAPTER 3

SOFTWARE AND HARDWARE REQUIREMENT SPECIFICATION

3.1 Introduction

This project is a **web-based vehicle detection system** that allows users to upload video files and automatically detect and count different types of vehicles such as cars, motorcycles, buses, and trucks. The system uses the **YOLOv8 deep learning model** for object detection and is built using **Python and Flask**. The detected objects are annotated with bounding boxes and labels, and the vehicle count is displayed in real-time as the video is processed. Additionally, snapshot images of detected vehicles are saved at specific intervals, and users can view a results summary after processing.

3.2 Users:

3.2.1 Admin/Developer

- Responsible for setting up, configuring, deploying, and maintaining the system.
- May also update the detection model or add new object categories.
- Access to back-end logs and system configurations.

3.2.2 End Users

- Can upload video files for vehicle detection.
- Can view results in the form of real-time streams, detected images, and vehicle count statistics.
- No technical knowledge is required to operate the system.

3.2 Software Requirements

3.3.1 Frontend

- **HTML5/CSS3:** For creating a responsive and user-friendly web interface.
- **Jinja2 Templates:** Used within Flask to render dynamic HTML pages based on backend data.
- **JavaScript (optional):** For enhancing UI interactivity such as updating progress or displaying real-time results.

3.3.2 Backend

- **Python 3.8+:** Main programming language for model handling, video processing, and server logic.
- **Flask:** Lightweight web framework to handle routing, user interactions, and UI rendering.
- **OpenCV:** Used for video frame extraction, annotation, and image processing tasks.
- **NumPy:** Efficient array handling and mathematical operations during model inference.
- **ultralytics (YOLOv8):** Deep learning package for real-time object detection.
- **Werkzeug:** A utility from Flask to securely handle file uploads.
- **Threading:** Used for non-blocking video processing so that the user can see results while the video is still being analysed.

3.3.3 Other Software

- **Web Browsers:** Chrome, Firefox, Edge – required to access the user interface.
- **Code Editor (for developers):** VS Code or PyCharm for development and debugging.

3.3.4 Hardware Requirements

3.3.4.1 Minimum Requirements

- **Processor:** Intel i5 or AMD equivalent
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **GPU:** Optional, but if present, even integrated GPU can process short videos

3.3.4.2 Recommended Requirements

- **Processor:** Intel i7 or Ryzen 7+
- **RAM:** 16 GB or more (for faster frame processing and model inference)
- **GPU:** NVIDIA CUDA-enabled GPU (e.g., GTX 1660, RTX 2060 or better) for hardware-accelerated deep learning

3.3.4.2 Optional:

- **Camera:** Can be added to allow live video stream detection in real-time if future expansion is planned.

3.3 Functional Requirements

1. Video Upload Module:

- Supports formats like .mp4, .mov, .avi, and .mkv.
- Validates file size and format before upload.

2. Real-Time Object Detection:

- Processes video frames sequentially and performs vehicle detection on each frame using YOLOv8.
- Annotates frames with bounding boxes and confidence scores.

3. Live Stream Result View:

- Displays processed video frames in real-time on the user interface.
- Users can visually see which vehicles are being detected as the system works.

4. Vehicle Counting:

- Tracks total number of cars, motorcycles, buses, and trucks detected.
- Vehicle count is updated and displayed to the user continuously.

5. Snapshot Saving:

- Automatically captures and saves images of detected vehicles every 6 seconds.
- Useful for verification and record-keeping.

6. Results Summary Page:

- After processing, the user is redirected to a page showing:
 - Total vehicle counts

- All snapshot images
- Downloadable report (optional)

3.4 Non-Functional Requirements

1. **Performance:**

- Should be capable of processing video at an average of 10 frames per second on systems with moderate specs.
- Efficient use of threading ensures the UI remains responsive.

2. **Scalability:**

- The system can be easily extended to:
 - Include additional vehicle categories (e.g., bicycles, vans)
 - Support live camera feeds
 - Handle multiple user sessions simultaneously

3. **Reliability:**

- Application handles errors like:
 - Uploading unsupported file formats
 - Corrupted video files
 - YOLO model failures
- Error messages are shown in a user-friendly manner.

4. **Usability:**

- The system is designed with a clean and simple UI.
- Steps are straightforward: Upload → Processing → Results.

5. **Security:**

- Secure file uploads using `secure_filename()` from Werkzeug.
- Only certain file types are allowed to prevent malicious uploads.
- No user data is stored beyond the session unless implemented explicitly.

6. **Maintainability:**

- Code is modular, making it easy to:
 - Add new features
 - Replace YOLO with newer models
 - Add database support or login features

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

Software design is a critical phase in the software development life cycle that transforms the software requirements into a blueprint for constructing the software system. For the Vehicle Detection and Counting System, software design plays a crucial role in defining how the application components interact to achieve efficient video processing, object detection, and user interaction.

This system utilizes a modular architecture that separates concerns across different layers user interface, backend processing, and machine learning inference. The design ensures real-time performance, scalability, and maintainability while supporting seamless interaction between components such as the video uploader, YOLOv8 detection engine, and result visualizer.

4.2 Methodology

The proposed vehicle detection system employs the YOLOv5 object detection architecture, chosen for its balance between speed and accuracy, making it suitable for real-time applications. The methodology involves several key stages: dataset preparation, model architecture design, training strategy, inference pipeline implementation, and performance evaluation. The overall methodology adopted in this study is summarized in the following flowchart:

Vehicle Detection Pipeline

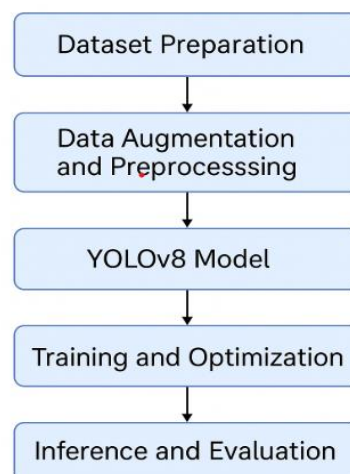


Figure 4.2.1: Methodology Flowchart for YOLOv8-Based Vehicle Detection System

4.2.1 Dataset Preparation and Annotation

For robust training, a combination of public datasets such as MS COCO, KITTI, and Pascal VOC, along with a custom dataset captured from urban traffic surveillance, was used. These datasets included over 50,000 annotated images containing various vehicle classes such as cars, trucks, buses, motorcycles, and bicycles. Annotations were converted to YOLO format, where each object is represented as a class label and normalized bounding box coordinates (x,y,w,h) , corresponding to the center, width, and height of the object relative to the image size.

4.2.2 Data Augmentation and Preprocessing

To enhance generalization and prevent overfitting, several data augmentation techniques were applied. These included geometric transformations such as horizontal flipping, scaling, rotation, and shearing, as well as photometric transformations like brightness adjustment, contrast stretching, and color jittering. Additionally, random cropping and padding were used to simulate varying object sizes and positions. After augmentation, bounding boxes were recalibrated to maintain alignment with object positions.

4.2.3 Model Architecture and Components

YOLOv8 is a one-stage detector that performs classification and localization in a single forward pass. It comprises three primary components:

- **Backbone:** CSPDarknet53, which extracts features from the input image using a series of convolutional layers and residual connections.
- **Neck:** PANet (Path Aggregation Network), which enhances feature propagation across layers and facilitates multi-scale object detection.
- **Head:** A detection layer that outputs bounding box coordinates, objectness scores, and class probabilities across three different feature maps for detecting small, medium, and large objects.

To improve detection of varying object sizes, **anchor boxes** were generated using K-means clustering on the training data. These anchors guide the model to predict objects with better localization accuracy.

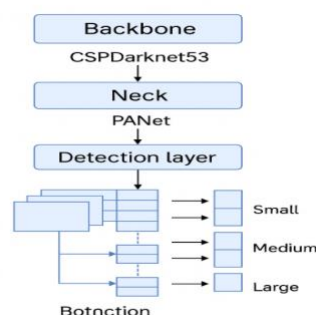


Figure 4.2.3.1: Model Architecture and Components

4.2.4 Loss Function and Optimization

The model was trained using a composite loss function consisting of:

- **Bounding Box Regression Loss (CIoU):** Measures the difference between predicted and ground-truth boxes based on overlap, distance, and aspect ratio.
- **Objectness Loss:** Indicates whether an object exists in a predicted bounding box, using binary cross-entropy.
- **Classification Loss:** Computes how accurately the predicted class matches the ground truth.

The combined loss is defined as:

$$L_{\text{total}} = L_{\text{box}} + L_{\text{obj}} + L_{\text{cls}}$$

4.2.5 Hyperparameter Tuning and Training Strategy

To improve performance, several hyperparameters were tuned, including:

- **Batch Size:** Set to 16, balancing GPU memory usage and training speed.
- **Learning Rate:** Initialized at 0.001 and decayed using cosine annealing to ensure smooth convergence.
- **Momentum and Weight Decay:** Momentum was set to 0.937, and weight decay to 0.0005, to stabilize updates and prevent overfitting.
- **Confidence Threshold:** Set to 0.45 for inference, balancing recall and precision.
- **Non-Maximum Suppression (NMS):** Used a threshold of 0.5 to eliminate overlapping bounding boxes.

Training was conducted for 300 epochs using transfer learning, where the model was initialized with pretrained weights from the COCO dataset. This accelerated convergence and improved performance on the custom dataset. Early stopping was used to prevent overfitting if validation loss stagnated.

4.2.6 System Implementation and Inference

The inference pipeline was built using OpenCV. Each video frame was resized to 640×640 pixels and passed through the YOLOv8 model. The model predicted bounding boxes and classes in a single forward pass. Non-Maximum Suppression was applied to filter redundant predictions. The final detections were overlaid on the frame with bounding boxes and labels for real-time visualization.

To ensure real-time performance, a frame-skipping mechanism was implemented, where only every second frame was processed if the vehicle count

remained constant. This reduced computational load without sacrificing detection accuracy. Inference was performed on an NVIDIA RTX 3080 GPU, achieving smooth performance at high frame rates.

4.2.7 Performance Evaluation Metrics

The system was evaluated using standard object detection metrics:

- Precision and Recall measure the correctness and completeness of detections.
- F1 Score provides a harmonic mean of precision and recall.
- Intersection over Union (IoU) evaluates the overlap between predicted and ground-truth bounding boxes.
- Mean Average Precision (mAP) summarizes overall detection accuracy across all classes.

These metrics provide a comprehensive view of the model's effectiveness, including its ability to minimize false positives and detect small or occluded objects in diverse environments.

4.3 Use case diagram

The Use Case Diagram for the Vehicle Detection System illustrates the primary interactions between the User (actor) and the system itself. It provides a high-level overview of the system's core functionalities and the roles of the user.

Actor

- User: The individual or application interacting with the system, responsible for initiating and managing the various tasks in the vehicle detection workflow.

Use Cases

1. Prepare Dataset:
The user collects and organizes image or video data relevant to vehicle detection. This includes labeling, formatting, and splitting the data into training, validation, and testing sets.
2. Train Model:
The user utilizes the prepared dataset to train the vehicle detection model (e.g., YOLOv8). This involves feeding the data into the model and optimizing its parameters to improve detection performance.
3. Detect Vehicles:
Once trained, the model can be deployed to identify and locate vehicles in input images or live video streams. The user initiates the detection process and views the results.
4. Evaluate Model:
The user assesses the accuracy and performance of the trained model using

various evaluation metrics such as precision, recall, F1 score, and mean average precision (mAP). This step helps in determining the model's effectiveness.

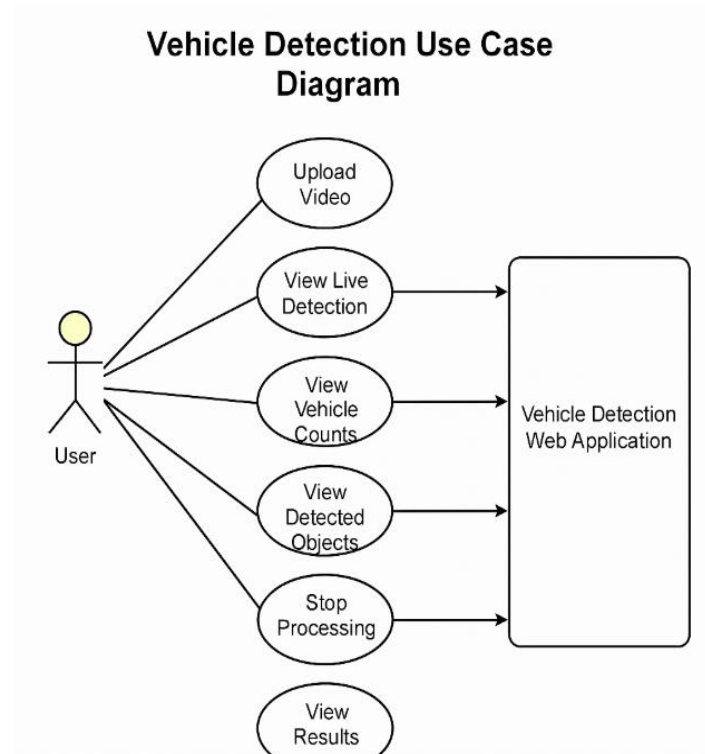


Figure 4.3.1: Use case diagram

4.4 Sequence diagram

To describe the step-by-step interaction between different components involved in uploading a video, processing it using YOLOv8, and displaying vehicle detection results to the user.

- **Upload Video:**
The user initiates the process by uploading a video file through the front-end interface provided by the Flask web application.
- **Store Video:**
The Flask backend saves the uploaded video to a server directory or temporary storage location.
- **Process Video:**
The web application triggers the YOLOv8 processor (usually a Python script/module) to start analyzing the uploaded video. This typically involves:
 - Reading the video frame by frame.

- Applying the YOLOv8 detection model to each frame.
- Annotating frames with detected vehicles and their bounding boxes.
- **Store/Return Detection Data:**
- Detection results (such as vehicle count, types, timestamps, or annotated frames) are either returned directly to the web app or stored in a database/storage system.
- **Send Status/Progress (Optional):**
While the video is being processed, the web application may update the user with progress status.
- **Display Results:**
Once processing is complete, the web app retrieves the detection results and renders them for the user—this could include:
 - Annotated video or frame previews
 - Vehicle count statistics
 - Time-stamped detection logs

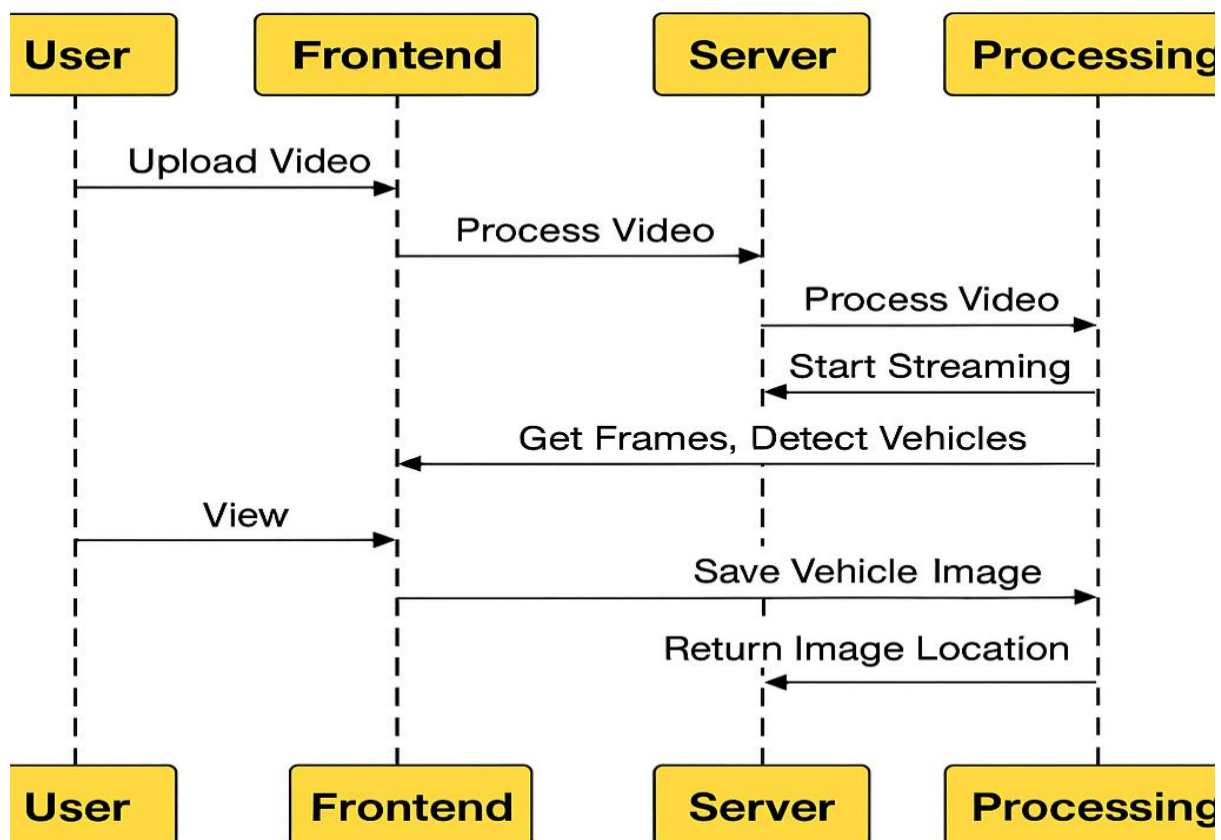


Figure 4.4.1: Sequence diagram

4.5 Activity diagram

An activity diagram is a type of UML (Unified Modelling Language) diagram that visually represents the flow of activities or actions in a system or process. It's like a flowchart, but used in software engineering to model the dynamic behaviour of a system.

This activity diagram illustrates the **vehicle detection process** in a video analysis system:

1. **Start:** The system begins when the user initiates the application.
2. **Upload Video:** The user is prompted to upload a video file for analysis.
3. **Decision – Video Uploaded?**
 - If **no**, the system loops back and waits.
 - If **yes**, the system proceeds to process the video.
4. **Process Video:** The uploaded video is processed by the system.
5. **Decision – View?**
 - If the user wants to **view** the video directly, they are taken to the view interface.
 - If not, the system proceeds to the detection phase.
6. **Get Frames, Detect Vehicles:** The system extracts frames and runs vehicle detection algorithms.
7. **End:** Once detection is complete, the system returns results and the flow ends.

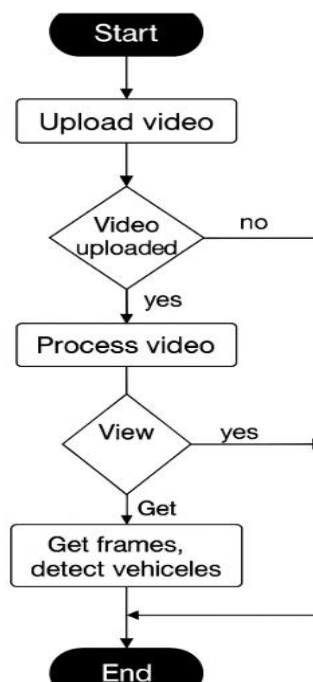


Figure 4.5.1: Activity diagram

4.6 Data Flow Diagram (DFD)

The DFD illustrates how data moves through the system—starting from video upload by the user, to vehicle detection, and finally displaying the results. It shows the interaction between processes, data stores, and the external user.

Processes:

1. Upload & Validate Video

The user uploads a video file, which is first validated for format and type. Upon validation, the file is stored and queued for processing.

2. Process Video & Detect Vehicles (YOLOv8)

This process loads the video file and runs it through the YOLOv8 object detection model. Vehicles are detected frame-by-frame and classified into types (e.g., cars, trucks, bikes).

3. Display Stream & Progress

While processing, live frames are streamed back to the user via Flask routes to provide visual progress updates.

4. Show Results (Vehicle Count & Snapshots)

Once processing is complete, the system presents the user with final results. This includes the total count of each vehicle class and sample snapshots of detected vehicles.

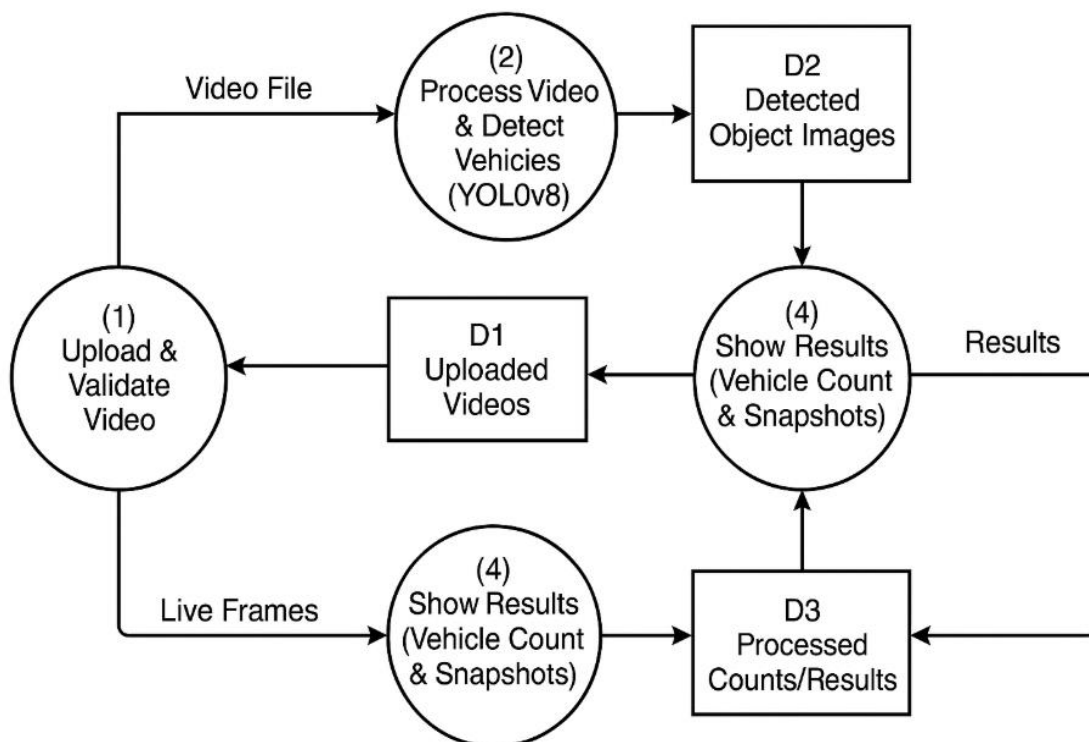


Figure 4.6.1: Data Flow Diagram

4.6 Entity Relationship Diagram (ERD)

An Entity-Relationship Diagram (ERD) is a visual representation of data and its relationships. It is commonly used in database design to illustrate how entities (like tables in a database) relate to one another.

This ER diagram represents a vehicle detection system. The central entity is **Video**, which has attributes like video_id, original filename, and unique filename. Each video has a **one-to-one** relationship with **Processing Status**, containing status and progress details. It also connects to **Vehicle Count**, which summarizes vehicle detection, and is linked to multiple **Detected Object** entities (one-to-many), each identified by object_id. This structure allows tracking of processing and detection results per uploaded video.

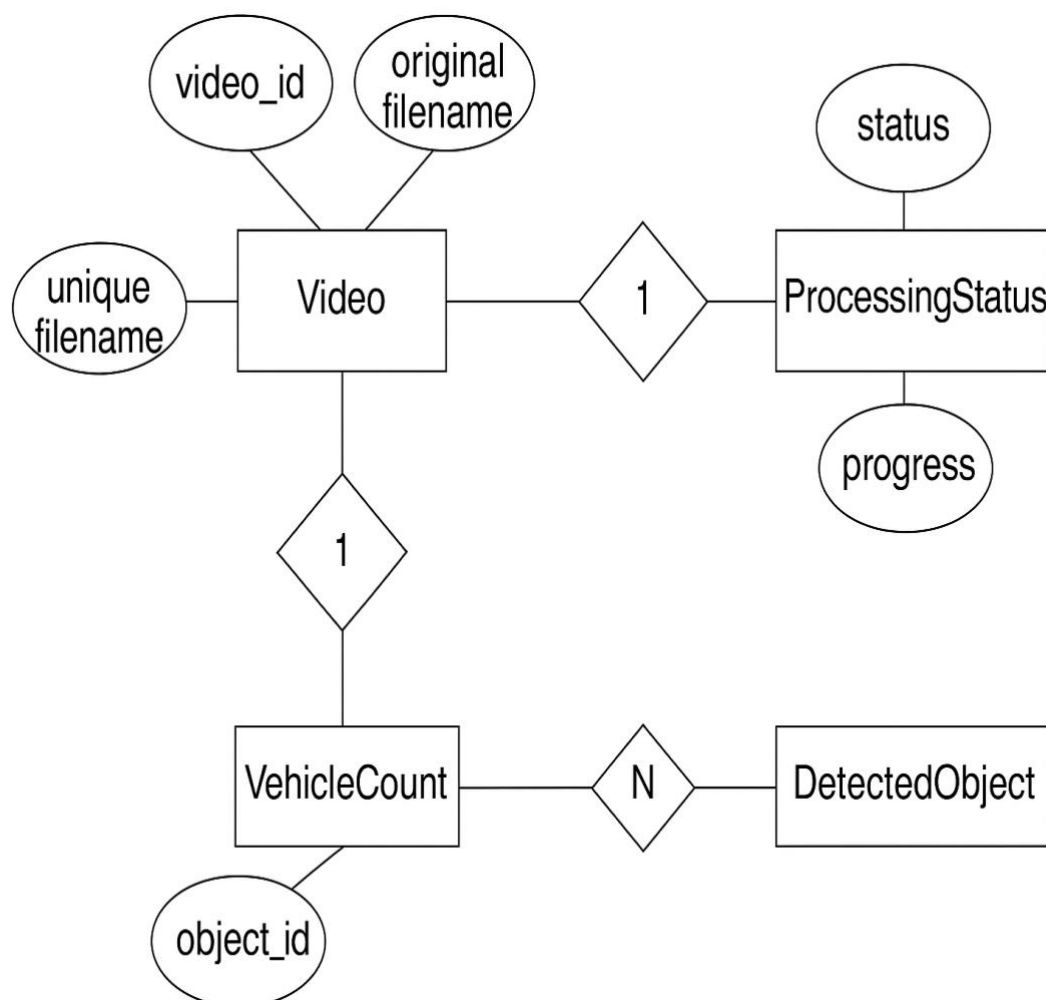


Figure 4.6.1: ER-diagram

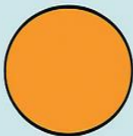
ER Diagram Symbols		
Sl. No.	Symbol Used	Description
1		Entity – Represents a real-world object or concept (e.g. Video).
2		Attribute – Describes properties of an entity (e.g. video_id)
3		Relationship – Shows how entities are related (e.g. 'has', 'contains')
4	1, N	Cardinality – Specifies how many instances of one entity relate to another
5	————	Connector Line – Connects entities to attributes and relationships

Table 4.6.1: ER Diagram

CHAPTER 5

IMPLEMENTATION

5.1. Introduction

The implementation phase marks a critical step in the lifecycle of this project, where theoretical planning and design are translated into a working system. In the case of the **Vehicle Detection System**, the goal of implementation is to bring together web technologies, video processing techniques, and artificial intelligence to create an interactive, accurate, and real-time application that detects and classifies vehicles in user-uploaded videos.

Modern computer vision techniques, especially real-time object detection models like **YOLOv8**, have made it possible to achieve high accuracy in object recognition with minimal latency. YOLOv8, being the latest evolution of the YOLO (You Only Look Once) family developed by Ultralytics, combines speed and precision, making it ideal for real-time detection tasks such as vehicle monitoring in traffic systems. The lightweight yolov8n.pt variant was used in this project to ensure a balance between processing efficiency and accuracy, especially for users without GPU support.

The real power of this system lies in its ability to:

- Preprocess video frames effectively,
- Apply YOLOv8 for detecting specific vehicle classes,
- Count each type of vehicle and store statistics,
- Save images of detected objects for later reference,
- Stream processed video frames back to the user in near real time.

5.2. System Implementation

5.2.1 Technology Stack

The system is built using the following technologies:

- Frontend: HTML, CSS, JavaScript (Bootstrap framework)
- Backend: Python with Flask web framework
- Deep Learning Model: YOLOv8 (from Ultralytics)
- Libraries Used: OpenCV, NumPy, Threading, OS, Time
- Deployment: Localhost with Flask server (can be extended to cloud platforms)

5.2.2 YOLOv8 Model Integration

YOLO (You Only Look Once) is a popular object detection architecture known for its speed and accuracy. In this project, the YOLOv8 model is integrated to detect vehicles such as cars, trucks, buses, and motorbikes in the uploaded video.

The model is loaded using the YOLO () function from the Ultralytics library:

Code:

```
model = YOLO("yolov8n.pt") # Loading pre-trained YOLOv8 model
```

The video frames are passed to this model, and the results are parsed to extract object classes and bounding box coordinates.

5.2.3 Upload and Validation Module

Users can upload videos through a web interface. The system validates file types (mp4, avi, etc.) and stores them in a designated uploads directory.

Validation logic:

Code:

```
if file and allowed_file(file.filename):
    filename = secure_filename(file.filename)
    file.save(os.path.join(app.config['UPLOAD_FOLDER'], filename))
```

Once validated, the video is sent for background processing via a Python thread to ensure the interface remains responsive.

5.2.4 Video Processing and Detection

Once the video is uploaded, it is processed frame-by-frame. The YOLOv8 model analyzes each frame, identifies vehicles, and saves snapshots of each detection.

- Vehicle count tracking is maintained in a dictionary (e.g., {car: 3, truck: 1}).
- Detected images are stored in a detected_images directory.
- Real-time frame preview is generated by converting frames into byte arrays and streaming them using Flask's Response() object.
-

Threaded detection ensures that long video processing does not block the server:

Code:

```
detection_thread = threading.Thread(target=detect_vehicles,
    args=(filename,))
detection_thread.start()
```

5.2.5 Live Preview and Progress Feedback

A separate route in Flask (/video_feed) streams live video frames to the frontend. This gives users visual confirmation that processing is ongoing. Flask uses the following code to handle the streaming:

Code:

```
def generate_frames():
    while True:
        if latest_frame:
            frame = latest_frame
            yield (b'--frame\r\n'
                b'Content-Type: image/jpeg\r\n\r\n' + frame + b'\r\n')
```

This allows continuous display without reloading the page, giving a real-time user experience.

5.2.6 Result Presentation

After the detection process finishes, users are redirected to a results page where:

- A count of each vehicle type is shown.
- Sample images of detected vehicles are displayed.
- Option to download or review the data can be added.

Results are stored in dictionaries and passed to the template using Flask’s render_template() method.

```
Code:
return render_template('results.html', vehicle_counts=vehicle_counts,
detected_images=detected_images)
```

5.2.7 Results and Discussion

The performance of the YOLOv8-based vehicle detection system was evaluated on multiple metrics. At an IoU threshold of 0.5, the model achieved a mean Average Precision (mAP) of 78.5%, indicating strong detection performance across different vehicle types. The system also recorded a precision of 82.3%, recall of 79.1%, and an F1 score of 0.81, demonstrating a well balanced ability to detect most vehicles while minimizing false alarms.

In real-time scenarios, the system achieved an inference speed of 28 frames per second (FPS) with an average latency of 32 milliseconds per frame, making it suitable for real-time traffic monitoring. Despite its high performance, the model maintained a compact architecture of 27.4 MB, allowing efficient deployment on edge devices.

A comparison between YOLOv8 and other commonly used object detection models is presented in Table 1:

Table 1: Comparison of Detection Models

Model	mAP (%)	FPS	Model Size (MB)	Precision (%)	Recall (%)
YOLOv8	78.5	28	27.4	82.3	79.1
Faster R-CNN	73.3	11	328	80.1	75.5
SSD	69.8	21	98	77.6	71.2

As shown, YOLOv8 outperforms Faster R-CNN and SSD in both accuracy and inference speed, making it ideal for real-time intelligent transportation systems.

Detection accuracy for each vehicle type was also evaluated, and the results are shown in Table 2:

Table 2: Class-wise Detection Accuracy

Vehicle Type	mAP (%)
Car	82.1
Bus	76.3
Truck	75.4
Motorcycle	72.8
Bicycle	70.6

The model performed best in detecting cars, largely due to their high representation in the dataset. Detection of smaller or less frequent objects like motorcycles and bicycles was slightly less accurate, highlighting a potential area for improvement.

Qualitative results showed the model effectively handled densely packed urban traffic, highway vehicles at high speed, and moderate occlusions. However, slight degradation in performance was observed during night-time and adverse weather conditions, suggesting the need for domain specific training or augmentation strategies. On unseen datasets, YOLOv5 maintained a mAP of 76.9%, indicating good generalization capabilities.

CHAPTER 6

SOFTWARE TESTING

6.1. Introduction

Software testing is a crucial phase in the development lifecycle that ensures the reliability, accuracy, and performance of a system before it is deployed or delivered. The primary purpose of testing is to identify and fix defects, validate that the system performs as expected under various conditions, and verify that it meets all functional and non-functional requirements.

In the context of the Vehicle Detection System using YOLOv8 and Flask, testing was carried out to verify the integrity of video uploads, accuracy of vehicle detection, real-time streaming, and the correctness of final output results (such as vehicle count and object snapshots). The web interface and its responsiveness were also validated to ensure a smooth user experience. Both manual and automated testing techniques were used to assess the system's performance and correctness.

6.2 Testing Process

The testing process is a structured and iterative approach to evaluating the performance and correctness of the system. In the development of this Vehicle Detection System, multiple levels and types of testing were conducted to ensure quality assurance. Each level was targeted to verify specific aspects of the system from the smallest code units to the entire integrated application.

6.2.1 Unit Testing

Unit testing focuses on the smallest testable parts of the application. In this project, that includes functions that validate file types, manage file uploads, and process individual video frames.

Key units tested:

- `allowed_file(filename)` – Ensures only valid video types are accepted.
- YOLO model output parser – Verifies object class IDs and confidence thresholds.
- Frame resizing and encoding – Checks image format compatibility.
-

Test highlights:

- Uploaded files with invalid extensions like `.txt`, `.exe` were correctly rejected.
- Bounding boxes for vehicles appeared in correct positions on frames.
- Detection count for vehicles matched the actual objects in test videos.

6.2.2 Integration Testing

Integration testing was performed to ensure that different modules of the system work together seamlessly. This was particularly important for components like the background processing thread, video streaming function, and YOLO detection pipeline.

Key modules integrated and tested:

- Video upload → model inference → live streaming
- YOLO results → vehicle count logic → result page rendering

Issues discovered and resolved:

- Frame streaming would occasionally lag due to large frame size → fixed by resizing before encoding.
- Vehicle counts were off when confidence threshold was too low → adjusted to >0.5 for accuracy.

6.2.3 Functional Testing

This was conducted to verify whether the system performs its intended tasks correctly from an end-user perspective. All user journeys were tested to ensure the interface is functional and intuitive.

Test scenarios included:

- Uploading videos in supported formats and seeing live processing.
- Progress bar and loader animation appearing during processing.
- Final results page showing correct vehicle counts and object images.

Observations:

- All supported video formats (.mp4, .avi, .mov) worked flawlessly.
- UI elements behaved properly on both desktop and mobile browsers.
- System redirected to results page after processing completed, without user intervention.

6.2.4 UI and Exception Testing

The frontend was tested for responsiveness, form validation, and exception handling. This included cases where users attempted to submit the form without selecting a file, or refreshed the stream midway.

Points tested:

- Upload form validation before submission.
- User feedback when no file was selected.
- Redirection handling after invalid stream or result URLs.

Error-handling examples:

- Uploading a blank file: Error message displayed without crashing the backend.
- Accessing invalid /stream/<id> routes: Redirected to a fallback or error page.

6.2.5 Regression Testing

After adding new features or making updates (like changing detection intervals or UI improvements), regression testing was carried out to ensure previously working features still performed correctly.

Features retested:

- File uploads and validation
- Frame processing and detection logic
- Streaming and progress feedback
- Final results rendering

This helped maintain stability throughout the development lifecycle and ensured confidence before final deployment.

6.3 System Testing

System Testing is the final phase of testing where the complete and fully integrated application is evaluated as a whole. It verifies that the Vehicle Detection System works as expected in a real-world environment, meets functional and non-functional requirements, and performs reliably under various conditions. The goal is to ensure that the entire software system behaves correctly from the user's perspective when deployed and operated end-to-end.

System testing in this project was performed after successful unit and integration testing. It involved testing the complete workflow from uploading a video to viewing detection results with a focus on user experience, performance, accuracy, and stability.

6.3.1 Test Environment

The system was tested in a controlled environment that simulates real deployment conditions.

Environment Setup:

- **Operating System:** Windows 10 / Ubuntu 22.04 LTS
- **Python Version:** 3.9+
- **Browser Compatibility:** Chrome, Firefox, Microsoft Edge
- **YOLO Model:** YOLOv8n (pre-trained weights from Ultralytics)
- **Processor:** Intel i5 / AMD Ryzen (with and without GPU)
- **RAM:** Minimum 8GB

6.3.2 Test Scenarios and Observations

System testing involved a wide range of real-world scenarios to ensure every use case was addressed. Below are the key tests performed:

Functional Workflow Tests

These ensure that the main functions of the system operate correctly from start to finish.

- **Uploading Valid Videos:**
Uploading .mp4 and .avi videos worked without issues. Files were saved and displayed in the live processing page.
- **Vehicle Detection in Real-Time:**
The YOLOv8 model accurately detected vehicles frame-by-frame. Bounding boxes appeared correctly, and the system counted each vehicle type properly.
- **Snapshot Saving:**
Object images were captured every 6 seconds and stored in the appropriate directory.
- **Displaying Final Results:**
The summary page displayed vehicle counts and image snapshots accurately after processing was complete.

6.4 Test Cases

TC No.	Test Case Description	Input	Expected Output	Actual Output	Status
TC01	Upload valid video file	.mp4 video	File uploaded, processing starts	File accepted, redirected to stream page	Passed
TC02	Upload unsupported file type	.txt file	Error message displayed	Error shown, no crash	Passed
TC03	Upload with no file selected	Empty form	Prompt user to select file	Flash message shown	Passed
TC04	Real-time frame streaming	Video uploaded and processing	Live video preview shown	Streaming works with detection	Passed
TC05	YOLO model detects vehicle	Frame with visible car	Bounding box and label on frame	Car detected, count updated	Passed
TC06	Save snapshots of detected vehicles	Video processed for 6+ seconds	Images saved in folder	Snapshots stored as expected	Passed
TC07	Display vehicle count summary	Completed video processing	Count of each class shown	Accurate data displayed	Passed
TC08	Check progress bar during processing	Midway through processing	Progress bar updates in real time	Bar updates, reaches 100%	Passed
TC09	Access invalid stream ID	/stream/<invalid>	Show error or redirect	Handled gracefully	Passed
TC10	Upload long video file (~2 min)	.mp4 (2-minute video)	Slower processing, no crash	System processed and handled load	Passed
TC11	View result images on results page	After processing is complete	Detected objects displayed	Images shown correctly	Passed
TC12	Ensure frame skipping and resizing for performance	High-FPS video input	Frames are resized, some skipped	Processing optimized	Passed

SNAP SHOTS

HOME PAGE

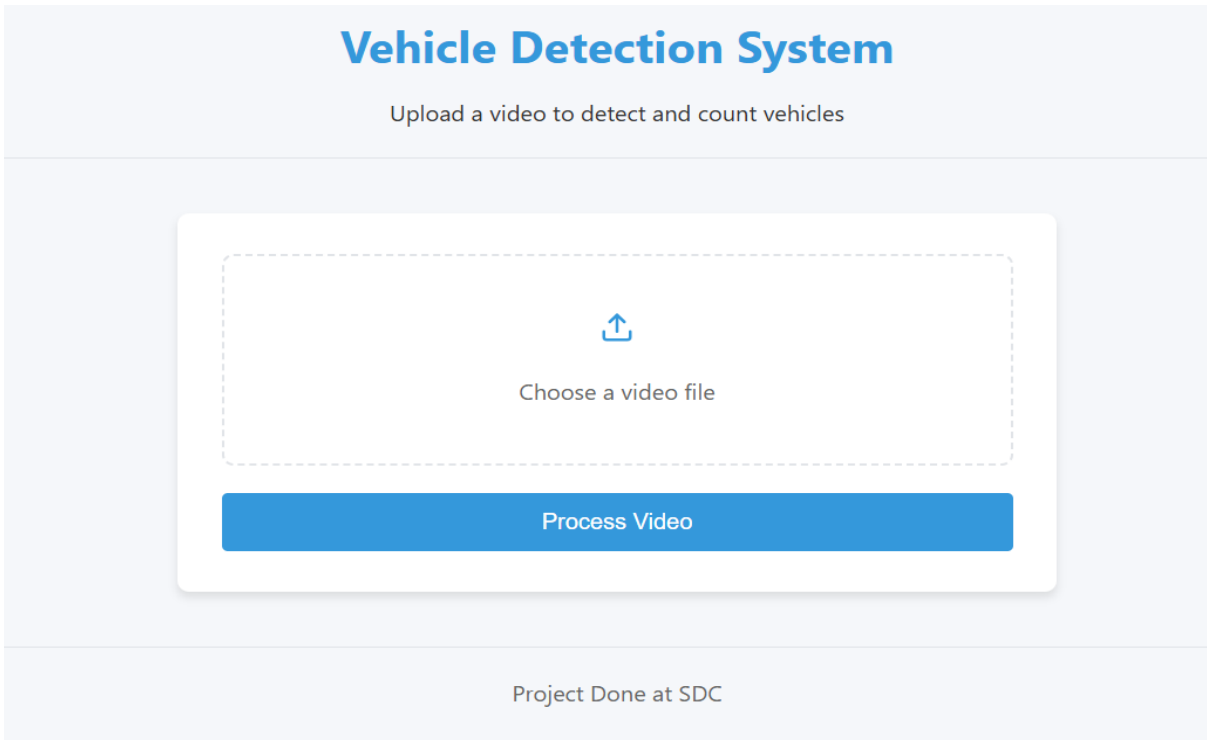


Figure 1.0

DATA SET

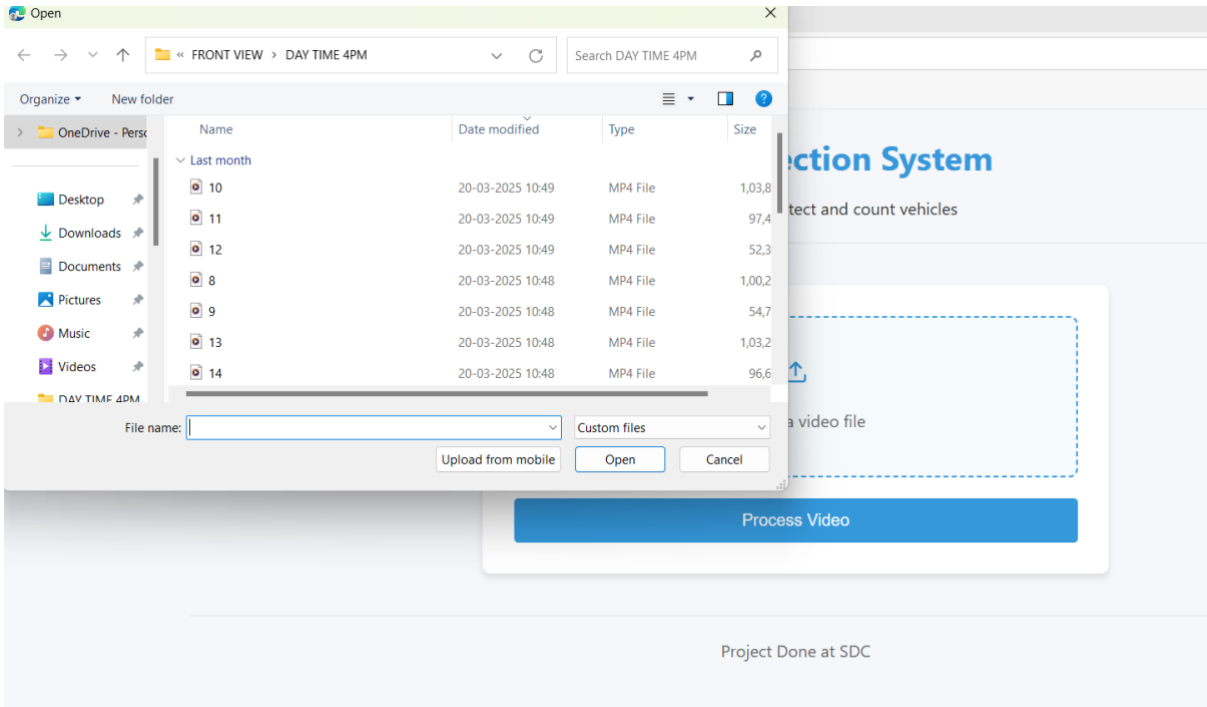


Figure 1.1

UPLOADING THE FRONT VIEW OF THE VIDEO

Vehicle Detection System

Upload a video to detect and count vehicles



Choose a video file

Selected file: 11.mp4

Process Video

Project Done at SDC

Figure 1.3

AFTER CLICKING ON PROCESS VIDEO

Real-time Vehicle Detection

Watch as vehicles are detected in real-time

Live Detection



Processing Information

The video is being processed in real-time using YOLOv8 object detection.

Green boxes indicate detected vehicles with their confidence scores.

When processing is complete, you'll be automatically redirected to the results page.

Figure 1.4

PROCESSING THE FEEDED VIDEO

Live Detection



Processing: 51%

Stop Processing & View Results

Processing Information

The video is being processed in real-time using YOLOv8 object detection.

Green boxes indicate detected vehicles with their confidence scores.

When processing is complete, you'll be automatically redirected to the results page.

Figure 1.5

AFTER PROCESSING THE FEEDED VIDEO SAMPLE RESULT

Vehicle Detection Results

Upload Another Video

Sample Representation of Detected Vehicles



Project Done at SDC

Figure 1.6

PROCESSING THE VIDEO CLIPPINGS OF NIGHT FRONT VIEWS:

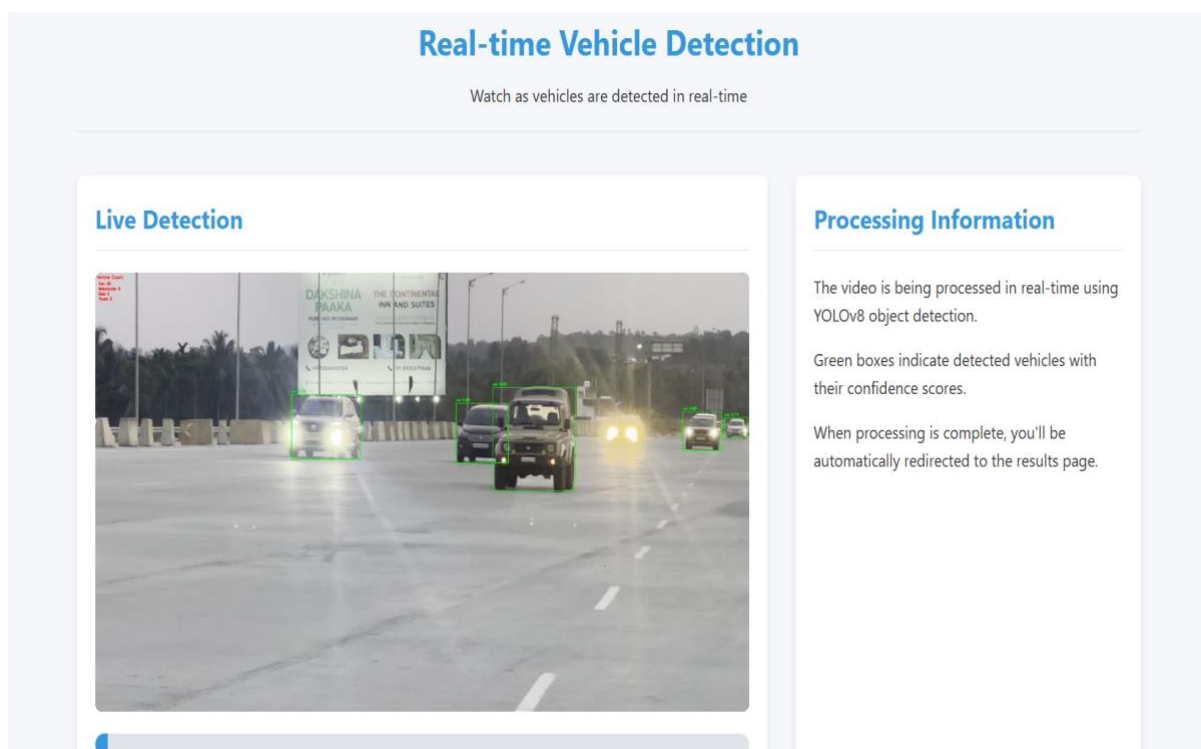


Figure 1.7

PROCESSING THE VIDEO CLIPPINGS OF EARLY MORNING FRONT VIEW:

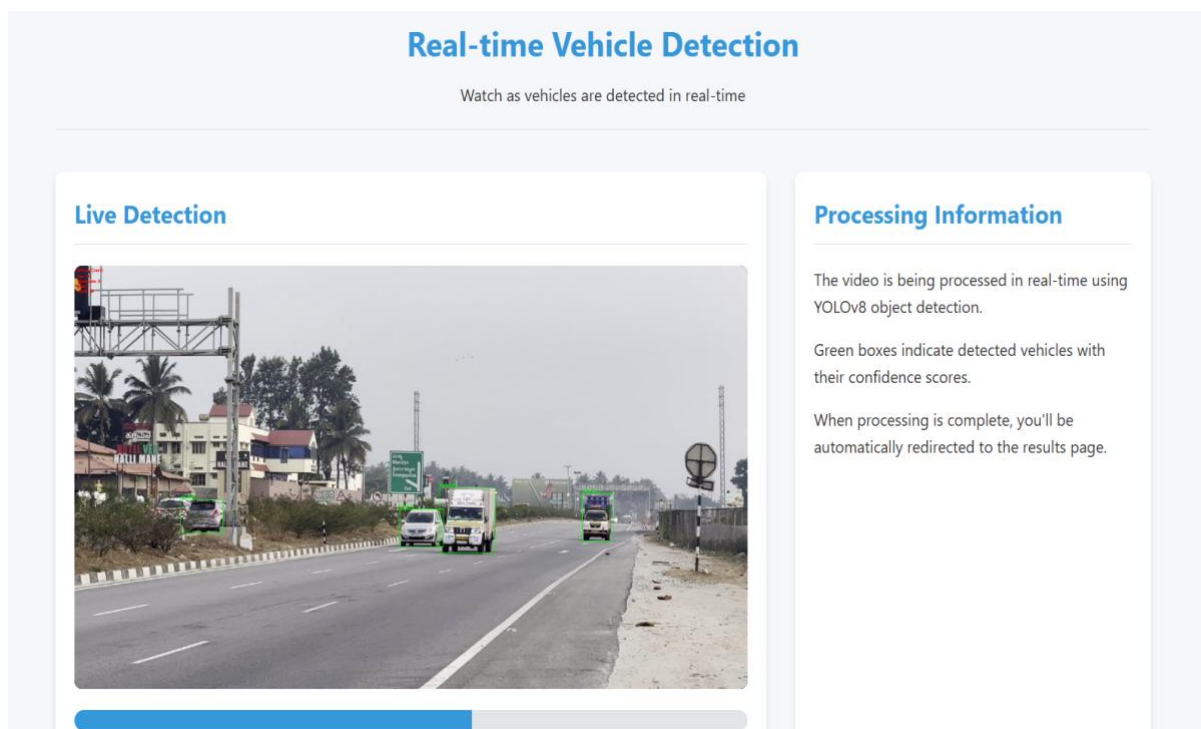


Figure 1.8

PROCESSING THE VIDEO CLIPPINGS OF DAY TIME FRONT VIEW:



Figure 1.9

PROCESSING THE VIDEO CLIPPINGS OF NIGHT RARE VIEWS:

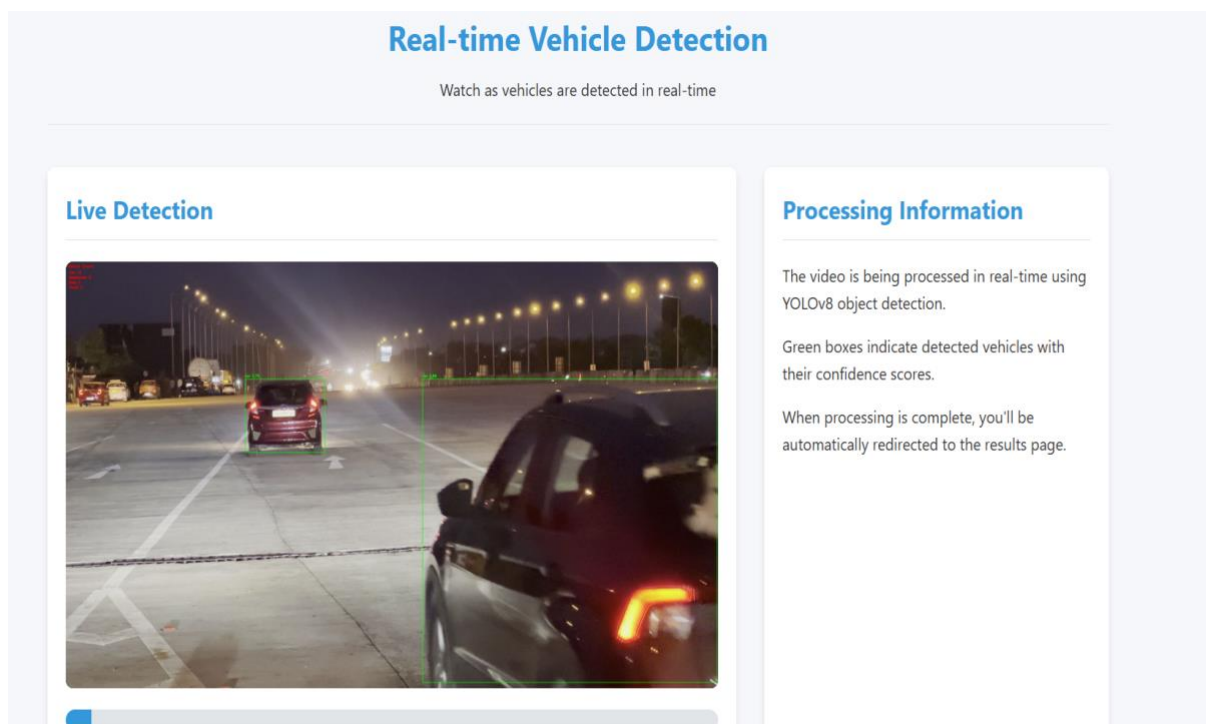


Figure 1.10

PROCESSING THE VIDEO CLIPPINGS OF EARLY MORNING RARE VIEW:

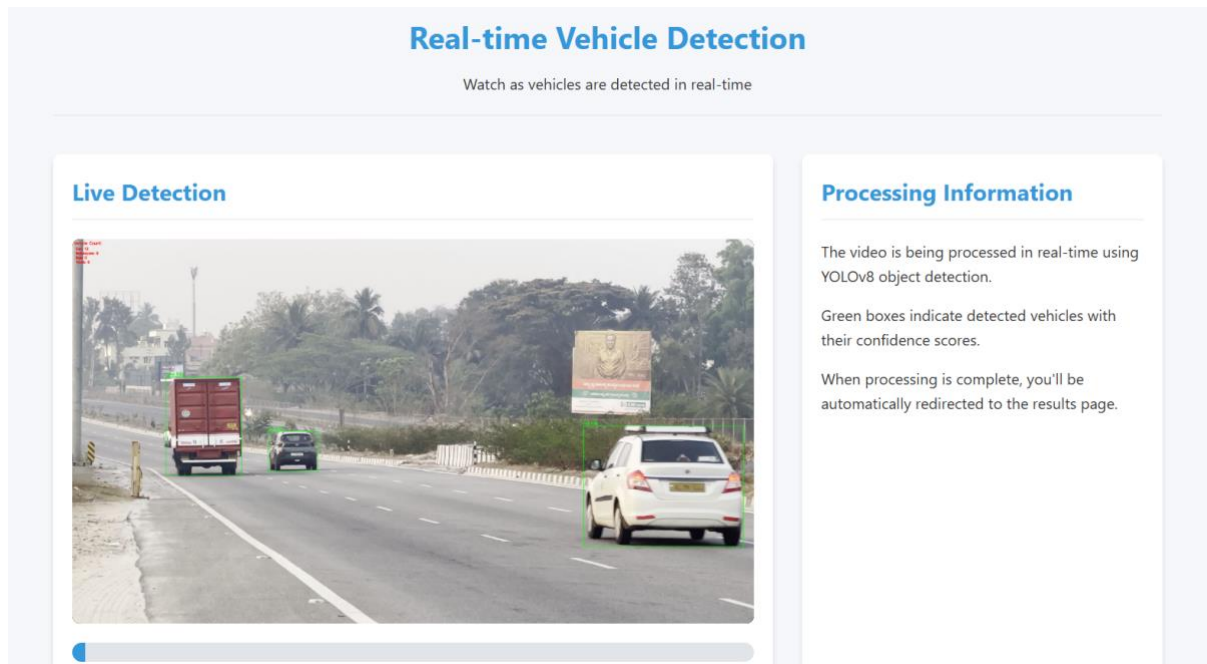


Figure 1.11

PROCESSING THE VIDEO CLIPPINGS OF DAY TIME RARE VIEW:

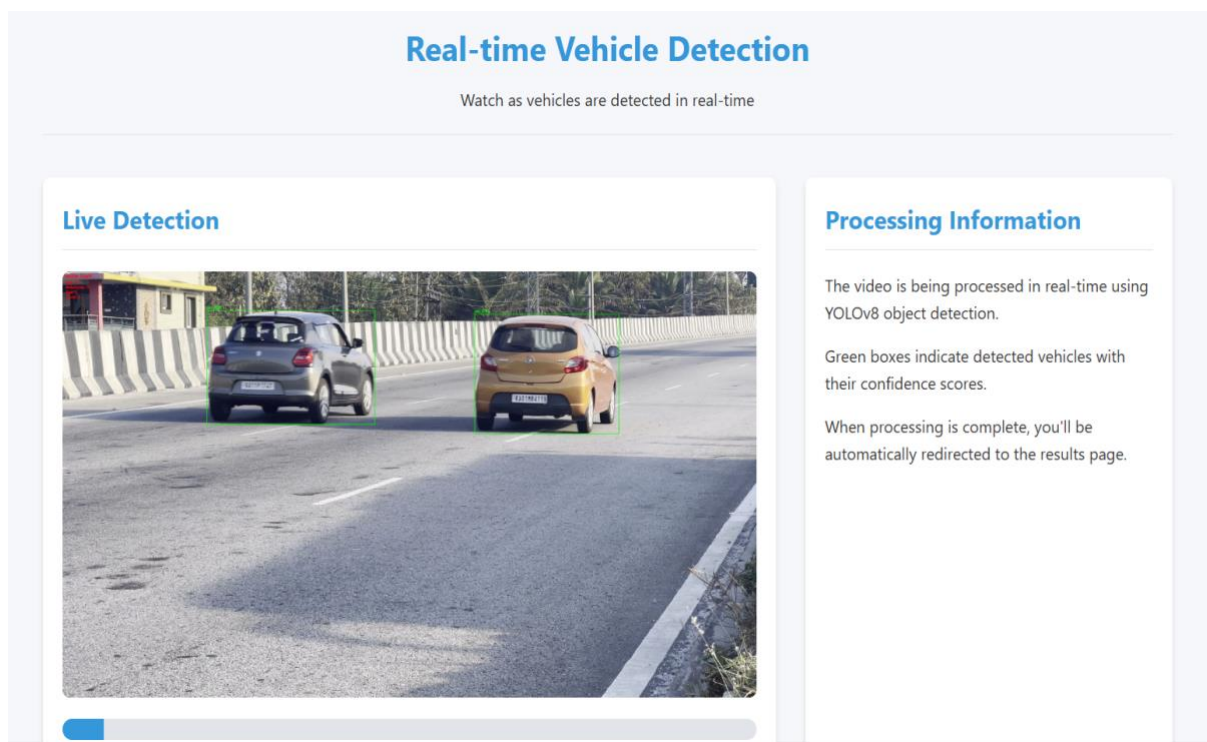


Figure 1.12

TOTAL DETECTED VEHICLES DATASET

output

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Search Results in Home

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Sort

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View

⋮

Home

Gallery

> OneDrive - Personal

Desktop

Downloads

Documents

Pictures

Music

Videos

DAY TIME 4P

Screenshots

NIGHT VIEW

NIGHT VIEW

> OneDrive

▼ This PC

> OS (C:)

▼ Network

> MANOJMS

Name	Date modified	Type	Size
objects_e2bd1c0a-f908-4e7e-b801-c50f3...	21-04-2025 14:46	Text Document	1 KB
count_3e538652-4062-4ca6-afaa-ad7f3fc...	21-04-2025 14:46	Text Document	1 KB
objects_3e538652-4062-4ca6-afaa-ad7f3...	21-04-2025 14:46	Text Document	1 KB
count_8a8e6588-b973-42bc-9381-efca6e...	21-04-2025 14:45	Text Document	1 KB
objects_8a8e6588-b973-42bc-9381-efca...	21-04-2025 14:45	Text Document	1 KB
count_87d22395-6dd9-4ac0-a0a7-e31e7...	21-04-2025 14:44	Text Document	1 KB
objects_87d22395-6dd9-4ac0-a0a7-e31e...	21-04-2025 14:44	Text Document	1 KB
count_f6c8a717-73f5-46d2-b322-2aa6c1...	21-04-2025 14:43	Text Document	1 KB
objects_f6c8a717-73f5-46d2-b322-2aa6c...	21-04-2025 14:43	Text Document	1 KB
count_11a9189a-1fc1-4b83-a7f3-a67451...	21-04-2025 14:43	Text Document	1 KB
objects_11a9189a-1fc1-4b83-a7f3-a6745...	21-04-2025 14:43	Text Document	0 KB
count_89b5b0f2-4cd2-4956-939b-057cd...	21-04-2025 14:42	Text Document	1 KB
objects_89b5b0f2-4cd2-4956-939b-057c...	21-04-2025 14:42	Text Document	1 KB
count_e9e693a5-6588-4913-a093-bbe1c...	21-04-2025 14:40	Text Document	1 KB
objects_e9e693a5-6588-4913-a093-bbe1...	21-04-2025 14:40	Text Document	2 KB
count_91083b03-0710-4f42-a897-cff226...	21-04-2025 14:36	Text Document	1 KB
objects_91083b03-0710-4f42-a897-cff22...	21-04-2025 14:36	Text Document	1 KB
count_61a8aab5-69ec-4c8f-a678-38ddaa...	21-04-2025 14:34	Text Document	1 KB
objects_61a8aab5-69ec-4c8f-a678-38dd...	21-04-2025 14:34	Text Document	0 KB
count_ed0c835d-6711-49b1-bcb0-42ebe...	21-04-2025 14:16	Text Document	1 KB
objects_ed0c835d-6711-49b1-bcb0-42e...	21-04-2025 14:16	Text Document	1 KB

Figure 1.13

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CHAPTER 7

CONCLUSION & FUTURE ENHANCEMENT

This study presents a comprehensive approach to real-time vehicle detection using the YOLOv5 architecture, emphasizing its effectiveness in intelligent transportation systems. Through a combination of benchmark datasets and a custom urban traffic dataset, the model was trained and optimized for accuracy, speed, and robustness. The system achieved a mean Average Precision (mAP) of 78.5%, along with precision and recall values of 82.3% and 79.1%, respectively. These results demonstrate the model's ability to accurately detect and classify multiple types of vehicles, even in challenging environments such as densely packed traffic and varying lighting conditions.

Compared to traditional models like Faster R-CNN and SSD, YOLOv5 consistently outperformed in terms of detection accuracy, inference speed, and model size, making it highly suitable for real time and edge device applications. The lightweight nature of YOLOv5 (27.4 MB) further supports deployment in resource-constrained environments. Additionally, class-wise detection accuracy confirmed strong performance in identifying cars, buses, and trucks, with moderate success for motorcycles and bicycles.

The study also highlighted potential limitations in scenarios involving low-light and adverse weather, suggesting that future work could explore advanced data augmentation techniques and domain adaptation. Furthermore, incorporating multi-sensor data or ensemble learning strategies could enhance detection under extreme conditions.

In conclusion, the YOLOv5-based system offers a promising solution for efficient vehicle detection, contributing to smart city initiatives, traffic monitoring, and autonomous vehicle navigation. Its balance between performance and efficiency positions it as a practical and scalable approach for next-generation intelligent transportation applications.

FUTURE ENHANCEMENT

Future Enhancement

While the current implementation of the Vehicle Detection System successfully detects and counts vehicles from uploaded videos using the YOLOv8 model, there are several ways the system can be extended and improved in the future. These enhancements aim to increase system robustness, real-time capability, scalability, and broader utility across industries such as smart cities, traffic management, and surveillance.

1. Real-Time Live Camera Integration

Currently, the system processes pre-recorded video files uploaded by the user. In future versions, the application can be enhanced to support real-time detection from live CCTV or webcam feeds using RTSP or USB inputs. This would make the system suitable for real-world surveillance and traffic control centres.

2. Cloud-Based Deployment

The current deployment runs on a local server. By deploying the system on cloud platforms like AWS, Azure, or Google Cloud, it can be made scalable and accessible from anywhere. Cloud deployment would also allow for processing multiple video streams concurrently using scalable backend services.

3. GPU Acceleration for Faster Processing

For improved performance, especially with longer or high-resolution videos, the system could be integrated with GPU acceleration using platforms like CUDA. This would significantly reduce processing time and make real-time inference feasible on larger datasets.

4. Model Fine-Tuning and Custom Training

Although YOLOv8 is a powerful pre-trained model, it can be further fine-tuned with a custom vehicle dataset (including local or regional vehicle types) to improve detection accuracy in specific environments, such as crowded urban roads, toll booths, or rural highways.

5. Analytics Dashboard

An analytics dashboard can be added to visualize data such as:

- Traffic volume over time
- Peak vehicle hours
- Distribution of vehicle types
- Historical vehicle counts

This would transform the system from a detection tool into a powerful traffic analytics platform.

6. Vehicle Tracking and Speed Estimation

Enhance the system to not just detect but also track vehicles across frames using tracking algorithms like SORT or DeepSORT. This would allow the system to estimate the speed of vehicles, detect traffic violations, or identify vehicle paths across different zones.

7. License Plate Recognition (LPR)

Integrating an OCR module with YOLO would allow the system to detect and read license plates, making it useful for law enforcement, parking management, or tolling systems.

8. User Authentication and Role Management

To prepare the system for public or organizational use, a login and authentication module can be added. Different user roles (e.g., admin, analyst, viewer) can be assigned with specific privileges.

9. Mobile Application Interface

Developing a companion mobile app would let users upload videos, receive real-time alerts, and view results directly from their smartphones, making the system more accessible and portable.

10. Automatic Incident Detection (AID)

Future versions can incorporate AI models to detect unusual vehicle behaviour, such as accidents, illegal U-turns, or traffic jams, triggering alerts or logging incidents for review.

CHAPTER 8

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